

VAYUNET

SIH IDEA Presentation — A Ready-to-Use Prompt for Every Slide

Built against the official SIH2025-IDEA-Presentation-Format.pptx structure

Paste each prompt into Claude, Antigravity, or any AI writing tool to generate that slide's content — already grounded in VayuNet's real research so the output isn't generic.

Before You Use These

- **The official template caps the deck at 6 slides total, including the title slide.** The reference sample you shared (CTRL+ALT+ELITE) added a 7th, unlabeled architecture-diagram slide — that breaks the stated limit. These prompts keep the diagram inside Slide 3 (Technical Approach), as the template itself instructs (“Methodology and process for implementation — Flow Charts/Images/working prototype”).
- Fill in the bracketed placeholders (Team ID, Team Name, member names) before using each prompt.
- Every prompt below tells the AI tool to output bullet points only, matching the official instruction to “avoid paragraphs and post your idea in points/diagrams/infographics.”
- The facts referenced in these prompts (datasets, competitor systems, architecture) come from VayuNet's own research documents — keep using those as your source of truth rather than letting the AI invent numbers when it fills in the slide.

Slide 1 — Title Page

This slide is pure metadata — no AI generation needed, but here's a prompt if you want it double-checked for completeness and formatting consistency.

PROMPT — Slide 1: Title Page

You are helping finalize the title slide of an SIH IDEA presentation.

Fill in this title slide exactly in this format, with no extra commentary:

Problem Statement ID - SIH26077

Problem Statement Title - Hyper-Local Severe-Weather Nowcasting Engine

Theme - Space Technology

PS Category - Software

Team ID - [YOUR TEAM ID]

Team Name - [YOUR TEAM NAME]

Check: does the Problem Statement Title match the exact wording used on the official SIH26077 listing? Flag if I should verify the exact official title text before submission.

Slide 2 — Idea Title / Proposed Solution

Covers: detailed explanation of the proposed solution, how it addresses the problem, and innovation/uniqueness — the three required fields on this slide.

PROMPT — Slide 2: Idea Title / Proposed Solution

You are a technical writer helping a Smart India Hackathon team fill in the "Proposed Solution" slide of their official IDEA presentation (SIH26077, VayuNet).

Write exactly three sections, in bullet points only (no paragraphs, 3-5 bullets each):

1. "Detailed explanation of the proposed solution"
2. "How it addresses the problem"
3. "Innovation and uniqueness of the solution"

Ground every claim in these real facts about VayuNet – do not invent new features or numbers:

- VayuNet predicts heavy-rainfall / severe-thunderstorm probability 2-6 hours ahead for a specific region, using a ConvLSTM-based spatiotemporal model.
- Inputs: INSAT-3D/3DR satellite channels (via MOSDAC), IndiaWeatherBench/IMDAA reanalysis (temperature, humidity, CAPE, CIN, IWV), and a Digital Elevation Model (CartoDEM/SRTM) for terrain.
- The model output is combined with DEM-based flow accumulation to flag which specific valleys are likely to flood, not just "will it rain."
- SHAP/Captum explainability translates the model's prediction into a plain-language causal chain (e.g. "moisture rising, atmospheric instability increasing").
- Existing systems it improves on: DGMR (DeepMind, radar-only, UK/US data), MetNet/MetNet-2 (satellite+radar, US-only, weights not public), GraphCast and Pangu-Weather (global medium-range, wrong scale for hyperlocal 0-6hr nowcasting). VayuNet is the first to combine India-specific satellite+reanalysis data with terrain-aware flood impact assessment at hyperlocal scale.
- Do not claim VayuNet invented AI nowcasting – the differentiation is the specific combination (India-specific data + terrain coupling + explainability), not the underlying AI technique.

Keep total text short enough to fit one slide alongside a diagram – aim for under 90 words per section.

Slide 3 — Technical Approach

Covers: technologies to be used, and methodology/process for implementation — include the architecture diagram here rather than as a separate slide, per the template's own instruction.

PROMPT — Slide 3: Technical Approach

You are a technical writer helping a Smart India Hackathon team fill in the "Technical Approach" slide of their official IDEA presentation (SIH26077, VayuNet).

Write exactly two sections in bullet points only:

1. "Technologies to be used"
2. "Methodology and process for implementation"

For section 1, list technologies grouped by category, based on VayuNet's real stack:

- ML/DL: PyTorch, PyTorch Lightning, Captum/SHAP for explainability
- Geospatial: xarray, rasterio, geopandas, richdem/pysheds (DEM flow accumulation)
- Backend: FastAPI
- Frontend: React + Leaflet (or Mapbox GL JS)
- Data sources: MOSDAC (INSAT-3D/3DR), IndiaWeatherBench/IMDAA reanalysis, CartoDEM/SRTM/Copernicus DEM
- Compute: Google Colab / Kaggle (T4/P100 GPUs) for training, ONNX for lightweight inference

For section 2, write a phase-based methodology using these real phases (condense to 4-5 bullets):

Phase 1 – data acquisition and preprocessing (reprojection, alignment, tiling)

Phase 2 – baseline models (persistence, optical flow, simple CNN) before the real model, to prove the ConvLSTM is actually adding value

Phase 3 – ConvLSTM backbone training with focal loss for rare-event class imbalance

Phase 4 – DEM-based flood-routing overlay + SHAP explainability integration

Phase 5 – FastAPI inference service + React dashboard integration

After the text, describe (in words, since I will draw it separately) a simple left-to-right architecture diagram: Data Sources (satellite + reanalysis + DEM) -> Preprocessing -> ConvLSTM Model -> Risk + Terrain Fusion -> FastAPI -> Dashboard/Alerts. Keep the diagram description to one sentence per box.

Keep it bullet-point only, no paragraphs, and make sure the whole slide's text is scannable in under 10 minutes of judge review time as required by SIH's own review process.

Slide 4 — Feasibility and Viability

Covers: analysis of feasibility, potential challenges and risks, and strategies for overcoming them — keep this one honest, since overclaiming feasibility is a common credibility failure.

PROMPT — Slide 4: Feasibility and Viability

You are a technical writer helping a Smart India Hackathon team fill in the "Feasibility and Viability" slide of their official IDEA presentation (SIH26077, VayuNet).

Write exactly three sections in bullet points only:

1. "Analysis of the feasibility of the idea"
2. "Potential challenges and risks"
3. "Strategies for overcoming these challenges"

Base this on VayuNet's real, verified constraints — do not soften or hide real limitations:

Real feasibility facts:

- MOSDAC, IndiaWeatherBench (Hugging Face), and CartoDEM/SRTM/Copernicus DEM are all free, registration-only, and accessible within 1-2 days.
- Google Colab Free (T4, 16GB VRAM) and Kaggle (P100, 16GB VRAM) are sufficient to train a regional-crop ConvLSTM backbone — this has been verified against current 2026 quota limits.
- Raw IMDAA reanalysis is too large/slow for a student team; IndiaWeatherBench is the practical ML-ready substitute.

Real challenges (do not omit these):

- IMD's Doppler Weather Radar only publishes rendered images publicly, not raw reflectivity data — radar cannot be a Phase-1 input.
- IMD's AWS/ARG ground-station portal was restricted to the public around May 2025 and access status should be re-verified before relying on it.
- Severe weather events (cloudbursts, flash floods) are rare, causing class imbalance in training data.
- A model trained on one region/season will not automatically generalize nationally or year-round.
- DEM-based flood routing is a first-order approximation, not a substitute for full hydrological survey.

Strategies (must map directly to the challenges above, not generic filler):

- Treat radar and live ground-station data as future/Phase-2 enhancements, not current dependencies.
- Use focal loss + temporal oversampling to address class imbalance.
- Scope the hackathon prototype explicitly to one region/season and state this limitation honestly rather than implying full coverage.
- Frame flood-routing output as "emergency response prioritization," not certified hydrology.

Keep language direct and specific — avoid vague terms like "scalability risks" without naming the actual risk and its real mitigation.

Slide 5 — Impact and Benefits

Covers: potential impact on the target audience, and benefits of the solution (social, economic, environmental, etc.)
— keep numbers realistic and clearly labeled as targets, not measured results.

PROMPT — Slide 5: Impact and Benefits

You are a technical writer helping a Smart India Hackathon team fill in the "Impact and Benefits" slide of their official IDEA presentation (SIH26077, VayuNet).

Write exactly two sections in bullet points only:

1. "Potential impact on the target audience"
2. "Benefits of the solution (social, economic, environmental, etc.)"

Ground this in VayuNet's real target audience and real mechanism — do not invent statistics:

Target audience: State Disaster Management Authorities, district administrations, IMD/MoES, communities in flood-prone hilly/riverine regions (e.g. Himalayan foothills, valleys prone to cloudburst-triggered flash floods like Kedarnath 2013, Chamoli 2021).

Real mechanism of impact:

- Converts a coarse 25km district-level forecast into a specific valley-level 2-6 hour warning.
- Gives emergency responders a terrain-prioritized list of at-risk zones (via DEM flow-routing), helping direct rescue resources before, not after, a flood.
- Plain-language causal explanations (via SHAP) make the warning usable by a non-meteorologist district official, not just a specialist.

For benefits, use these categories honestly:

- Social: more precise, more actionable severe-weather warnings for vulnerable hill/valley communities specifically (not "everyone in India" — be specific about who benefits most).
- Economic: potential to reduce disaster-response costs by improving lead time for evacuation/resource positioning (frame as a hypothesis to validate, not a guaranteed dollar figure).
- Environmental / public health: not a strong angle for a weather nowcasting tool — skip or keep minimal rather than manufacturing an artificial environmental benefit.
- Government relevance: built entirely on open MoES/ISRO data, with a plausible pilot path with IMD or a State DMA.

Any numeric target (e.g. "% reduction in response time") must be labeled explicitly as a target pending real-world pilot validation, not a measured outcome — do not present it as an already-achieved result.

Slide 6 — Research and References

This is a links-only slide. The prompt below asks the AI to format a clean reference list rather than generate new content — the sources themselves come from VayuNet's own verified research.

PROMPT — Slide 6: Research and References

You are formatting the "Research and References" slide of an SIH IDEA presentation (SIH26077, VayuNet). This slide should contain only a clean list of real, working links – no invented sources.

Format the following verified sources as a clean bullet list, grouped under short subheadings (Data Sources / Reference Systems / Standards), one link per line, no extra commentary:

Data Sources:

- <https://www.mosdac.gov.in> (INSAT-3D/3DR/3DS satellite data, ISRO/MOSDAC)
- <https://huggingface.co/datasets/tungnd/IndiaWeatherBench> (ML-ready IMDAA reanalysis subset)
- <https://cds.climate.copernicus.eu> (ERA5 reanalysis, ECMWF/Copernicus)
- <https://bhuvan.nrsc.gov.in> (CartoDEM terrain data, ISRO/NRSC)
- <https://mausam.imd.gov.in> (IMD public weather/radar reference)

Reference Systems (for competitive/technical grounding):

- https://github.com/openclimatefix/skillful_nowcasting (DGMR, DeepMind's radar nowcasting model)
- <https://github.com/google-deepmind/graphcast> (GraphCast, global medium-range forecasting)
- <https://github.com/198808xc/Pangu-Weather> (Pangu-Weather, global medium-range forecasting)

Do not add sources I have not explicitly listed here – if you believe an additional citation would strengthen this slide, name it separately as a suggestion rather than inserting it directly into the list.

Reminder: after generating all six slides, run the official checklist yourself — six slides maximum including title, bullet points only, one saved-as-PDF file for portal upload, and no deviation from the section headings the template specifies.